

Quantum Computing Approaches to Vehicle Routing Problem and its variants

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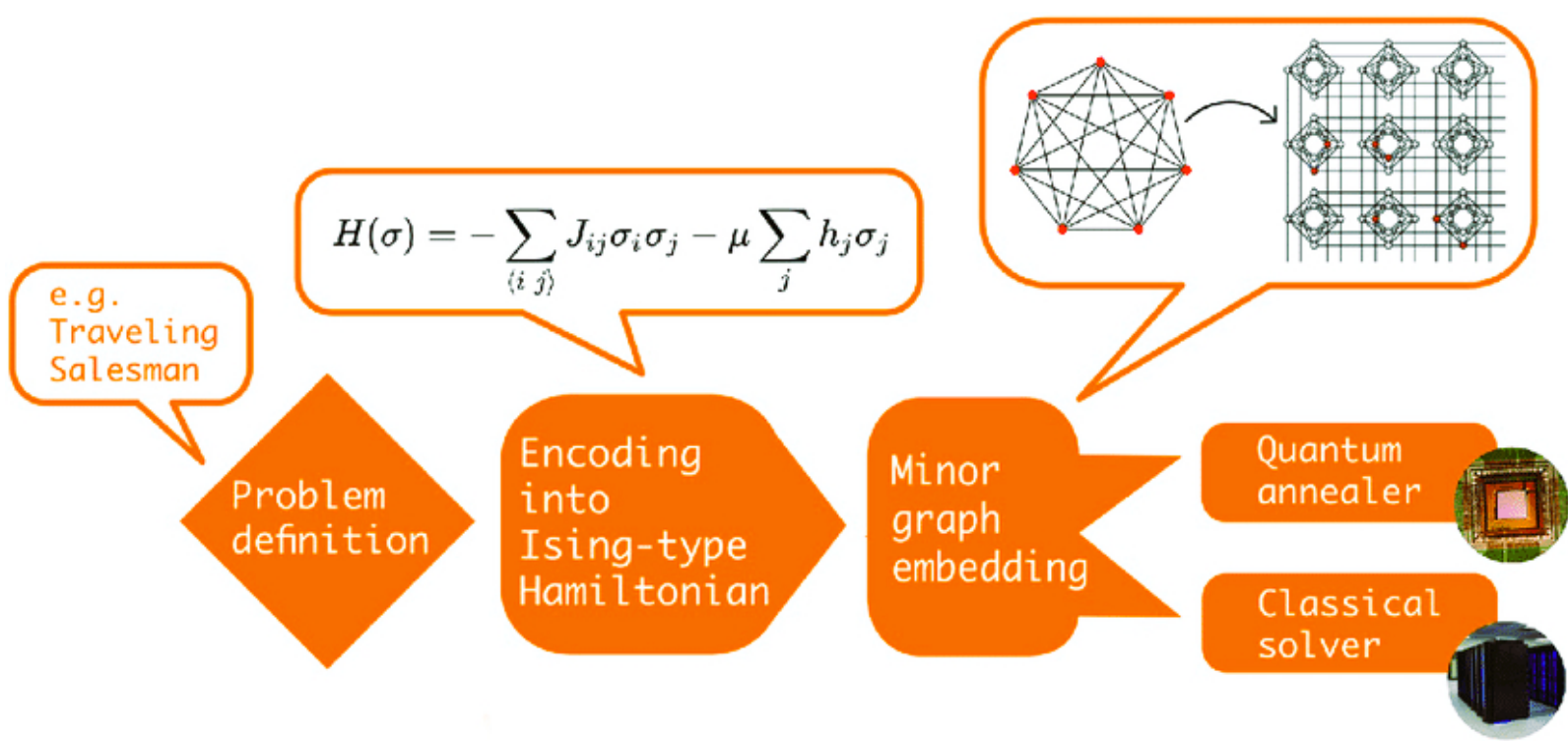
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Motivation

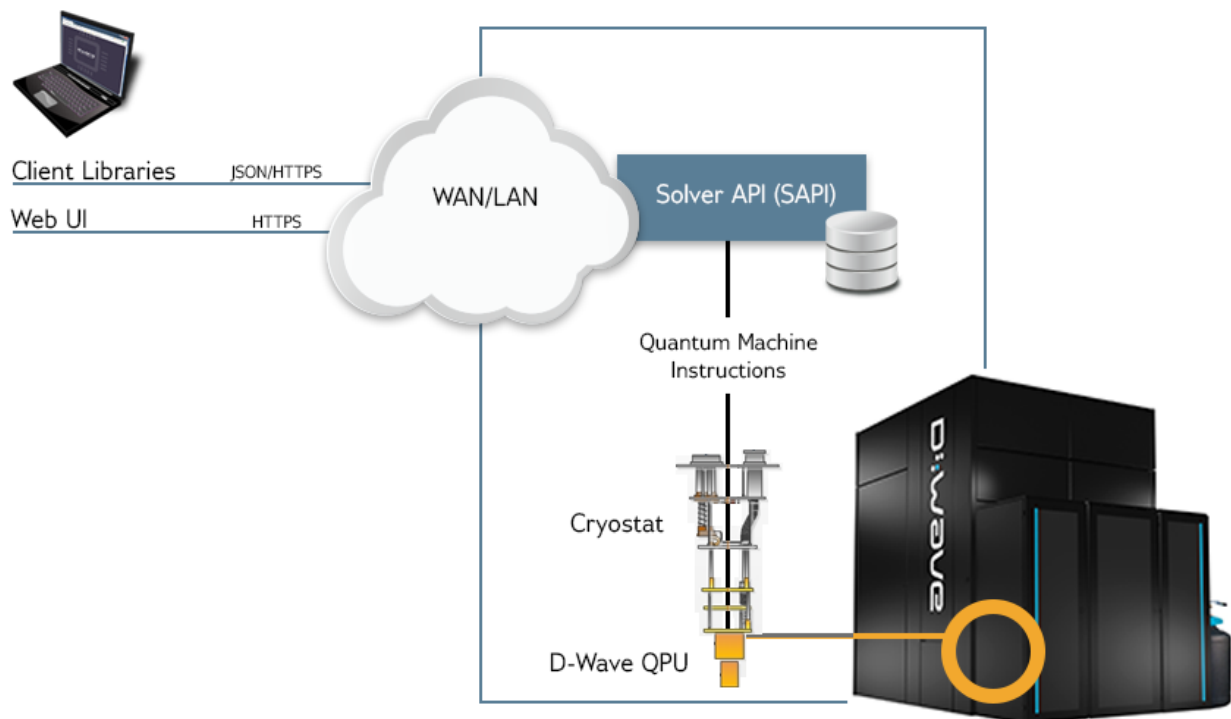
- **Complexity of VRP** – VRP’s intricate nature in optimizing vehicle routes demands innovative solutions beyond classical methods.
 - **Quantum Annealing’s Parallelism** – Leverage quantum annealing’s [parallel processing to explore multiple routes simultaneously](#), accelerating solution discovery.
 - **Continuous Quantum Models’ Flexibility** – CQMs offer a continuous framework for adapting to the dynamic and continuous nature of real-world vehicle routing scenarios.
 - **Challenges in Traditional Approaches** – Traditional methods often struggle with the dynamic and combinatorial aspects of VRP, necessitating a quantum computing paradigm.
 - **Faster Convergence and Scalability** – Quantum approaches [promise faster convergence to optimal solutions](#) and scalability to handle larger instances of the VRP.
- Real-World Impact** – Envisioning tangible benefits in reduced operational costs, minimized environmental impact, and enhanced logistics efficiency through optimized vehicle routes.
- **Contribution to Quantum Computing Applications** – Contributing to the expanding field of quantum computing applications by applying novel approaches to address practical logistics challenges.
 - **Future Vision** – Striving towards a future where quantum computing transforms the landscape of vehicle routing, offering sustainable and efficient solutions for logistics and transportation.

Quantum Annealing and Adiabatic Quantum Computing

Typical Workflow



D-Wave



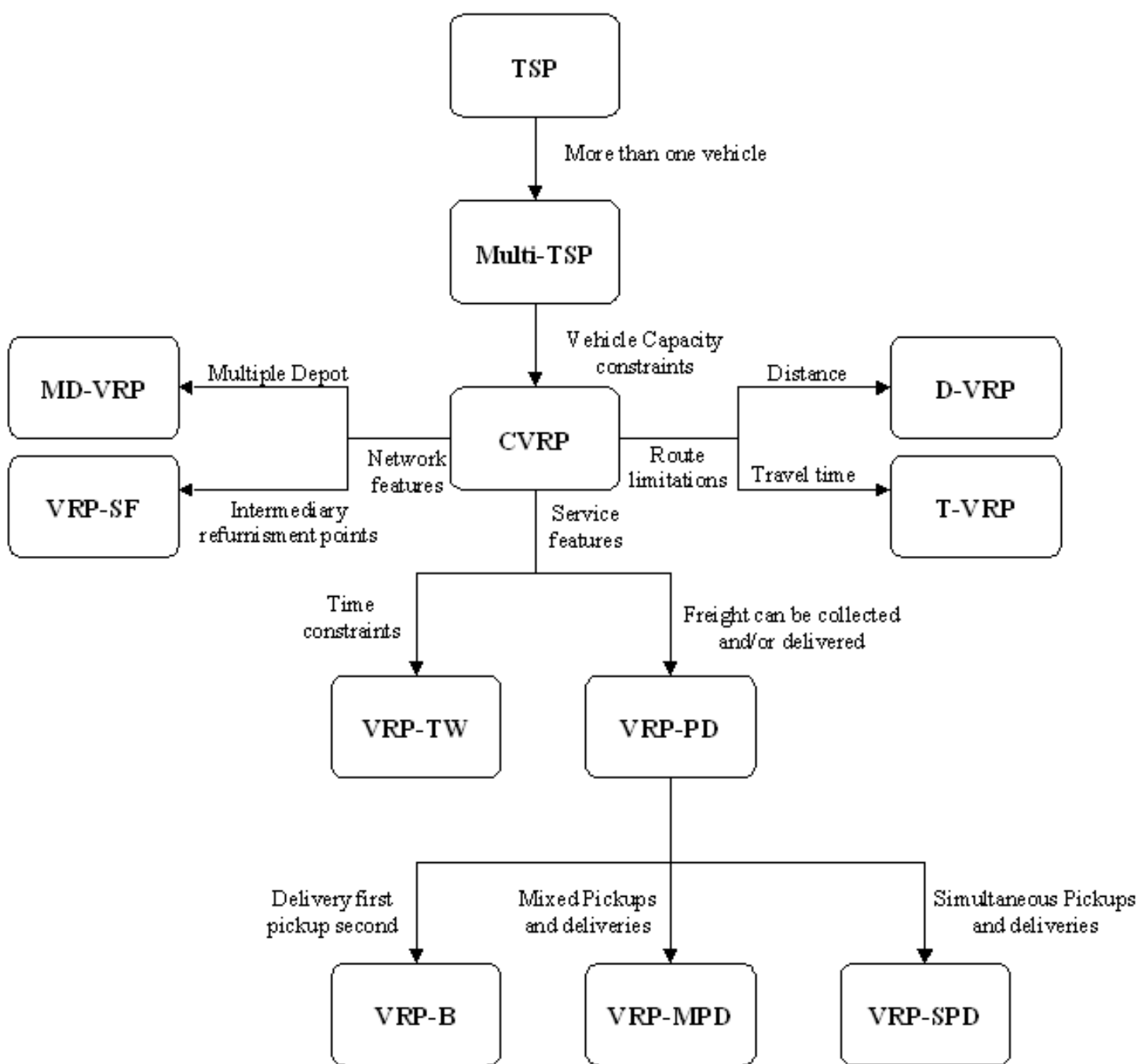
Constrained Quadratic Model



Vehicle Routing Problem

VRP is a [combinatorial optimization challenge integral to logistics](#), involving the efficient routing of vehicles to service a set of locations. The core objective is to minimize travel costs while satisfying constraints, making it pivotal for streamlined transportation and resource utilization.

Variants of VRP



Research objectives

Our research focuses on the following

- **Quantum Annealing Evaluation:**
 - Assess D-Wave’s quantum annealing for various VRP variants.
 - Measure [scalability and efficiency across different VRP complexities](#)
- **Constrained Quadratic Model (CQM) Analysis:**
 - Explore [CQMs for optimizing routes in diverse VRP scenarios](#).
 - Analyze CQMs versus traditional quantum models in different VRP contexts.
- **Comparison with Classical Approaches**
 - [Contrast quantum annealing and CQM with classical algorithms for VRP](#).
 - Evaluate computational efficiency and solution quality in addressing variant-specific challenges.

VRP Variant under consideration

If each node in the Vehicle Routing Problem (VRP) can be visited by more than one vehicle variant, this variant is commonly referred to as the [Multiple Vehicle Visits \(MVV\) variant](#) or [Node Visits by Multiple Vehicles \(NVMV\)](#). This variation allows for increased flexibility in the assignment of vehicles to specific nodes, potentially leading to more efficient and dynamic routing solutions.

Formulation

- **Decision Variables:** The decision variable $x_{k,i,j}$ represents [whether vehicle \$k\$ travels from node \$i\$ to node \$j\$](#) . It is a binary variable that takes the value 1 if the vehicle travels from i to j and 0 otherwise.

$$x_{k,i,j} \in \{0,1\} \quad \forall k \in [0, K-1], i, j \in [0, n-1]$$

- **Objective Function:** The objective function aims to minimize the total cost of vehicle routes. It is the sum of the product of the distance between nodes i and j ($W_{\text{adj}[i][j]}$) and the binary decision variable $x_{k,i,j}$ over all vehicles and nodes.

$$\text{Minimize: } \sum_{k=0}^{K-1} \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} W_{\text{adj}[i][j]} \cdot x_{k,i,j}$$

- **Constraints:**
 - **Outgoing Edges from Depot:** Ensure that each vehicle leaves the depot.

$$\sum_{k=0}^{K-1} x_{k,0,j} = 1 \quad \forall j \in [1, n-1]$$

- **Vehicle Visits to Each Node:** Each node should be visited by at most one vehicle.

$$\sum_{i=0}^{n-1} x_{k,i,j} \leq 1 \quad \forall k \in [0, K-1], j \in [0, n-1]$$

- **Flow Conservation:** [Flow into and out of each node must be balanced for each vehicle](#).

$$\sum_{i=0}^{n-1} x_{k,i,j} - \sum_{i=0}^{n-1} x_{k,j,i} = 0 \quad \forall k \in [0, K-1], j \in [0, n-1]$$

- **Each Node is Visited at Least Once:** Ensure that each node is visited at least once.

$$\sum_{k=0}^{K-1} \sum_{i=0}^{n-1} x_{k,i,j} \geq 1 \quad \forall j \in [1, n-1]$$

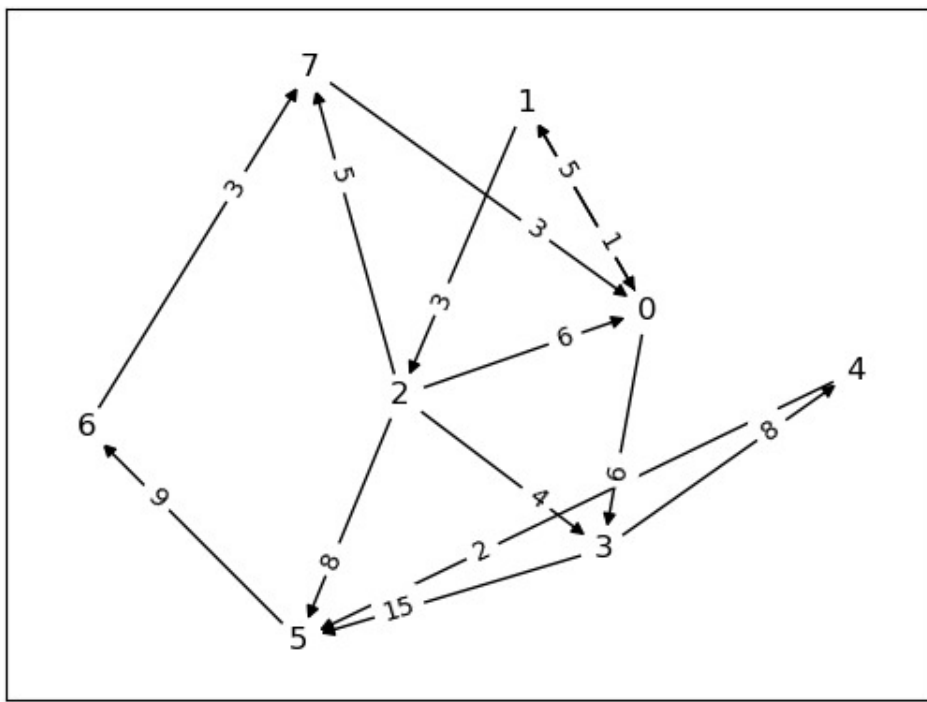
- **Sub-tour Elimination:** Prevent sub-tours by [introducing auxiliary variables \$u_{k,i}\$ and constraints](#).

$$u_{k,i} - u_{k,j} + Q \cdot x_{k,i,j} \leq Q - 1 \quad \forall k \in [0, K-1], i, j \in [0, n-1], W_{\text{adj}[i][j]} \neq 0, j \neq 0$$

Here, Q is a sufficiently large constant.

Experiments and Results

Input VRP:



Comparison of QPU and LP Runtimes:

Solver	QPU Access Time	Charge Time	Run Time
CQM	00 h : 00 m : 00.032 s	00 h : 01 m : 00.000 s	00 h : 01 m : 00.432 s
LP	N/A	N/A	00 h : 00 m : 00.056 s

References